

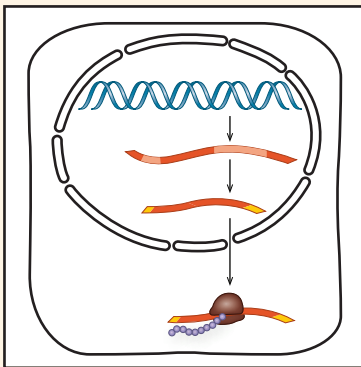
17 Gene Expression: From Gene to Protein

KEY CONCEPTS

- 17.1** Genes specify proteins via transcription and translation p. 336
- 17.2** Transcription is the DNA-directed synthesis of RNA: *A Closer Look* p. 342
- 17.3** Eukaryotic cells modify RNA after transcription p. 345
- 17.4** Translation is the RNA-directed synthesis of a polypeptide: *A Closer Look* p. 347
- 17.5** Mutations of one or a few nucleotides can affect protein structure and function p. 357

Study Tip

Make a visual study guide: Sketch the process shown below, and add labels and details as you read the chapter. (In this exercise, assume all processes take place in a eukaryotic cell.)



Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 17
- BioFlix® Animation: Protein Synthesis
- Figure 17.27 Walkthrough: Types of Small-Scale Mutations that Affect mRNA Sequence

For Instructors to Assign (in Item Library)

- BioFlix® Tutorial: Protein Synthesis (1 of 3): Overview
- Tutorial: CRISPR: A Revolution in Genome Editing

Ready-to-Go Teaching Module
(in Instructor Resources)

- Gene Expression: Mutations (Concept 17.5)



Figure 17.1 A population of albino donkeys grazes on vegetation on the hillsides of Asinara, an Italian island. Several centuries ago, a recessive mutation that disables pigment synthesis arose in the DNA of one donkey and was passed down through the generations. Inbreeding has resulted in a large number of homozygous albino donkeys living on the island today.

How can one change in DNA result in such a dramatic change in appearance?

Proteins are the link between genotype and phenotype. Gene expression is the process by which DNA directs the synthesis of proteins:

